This document presents a detailed analysis of web analytics data, focusing on relationships among visits, unique pageviews, exits, bounces, and time spent on pages, along with the influence of source groups and continents.

- **Visits strongly predict unique pageviews:** There is a strong positive correlation ($r = 0.8144$, $p < 0.001$) between visits and unique pageviews, supported by a linear regression model explaining 66.33% of the variance, indicating that as visits increase, unique pageviews also increase significantly.
- **Exits influenced by multiple factors:** Exits are strongly affected by bounces, unique pageviews, visits, time on page, and certain source groups, with bounces and unique pageviews having the most significant positive effects. The model accounts for 79.07% of the variation in exits, while continent variables mostly show no significant effect.
- **Bounce rates relate to source and engagement metrics:** Bounce rates are significantly impacted by source group, time on page, unique pageviews, visits, and exits. Time on page and unique pageviews negatively affect bounces, while visits and exits have positive effects. The model explains 78.55% of bounce variation.
- **Time on page affected by several variables but with weak model fit:** Time on page is significantly influenced by bounces (negative effect), visits, unique pageviews, exits, and the South America continent, though the model explains only about 9.66% of its variation, indicating other factors may be involved.

Web analytics screenshots

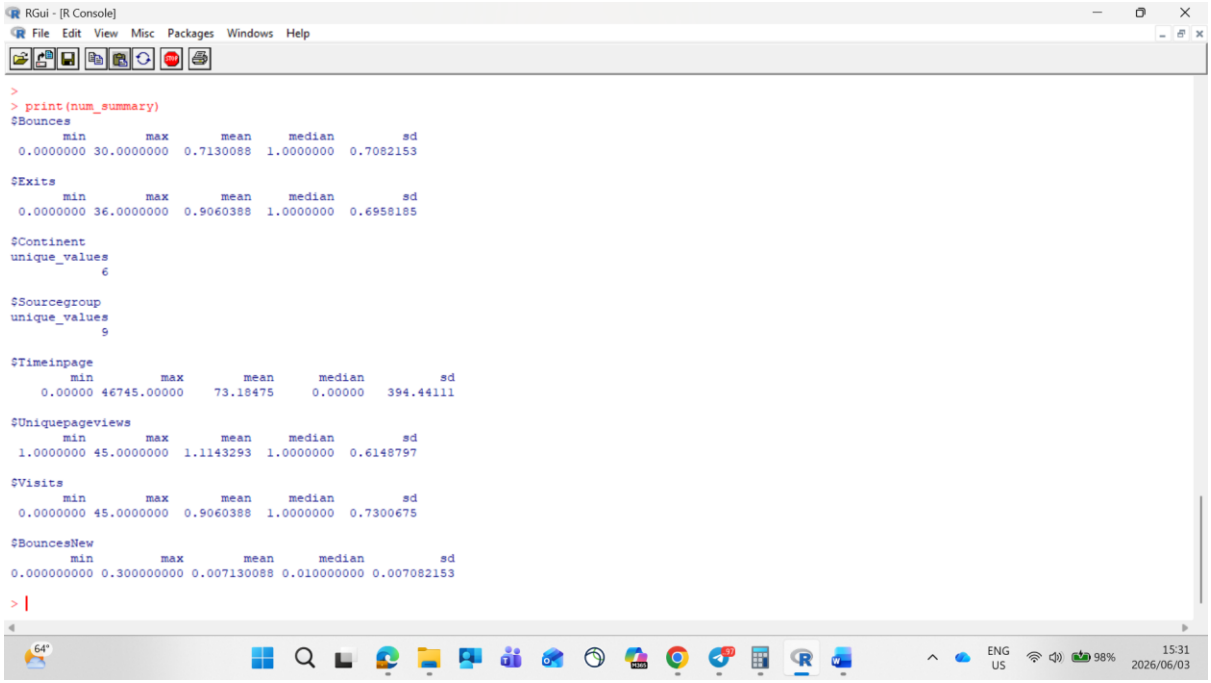
Data structure

```
tibble [32,109 × 8] (S3: tbl_df/tbl/data.frame)
 $ Bounces        : num [1:32109] 0 0 0 0 0 0 0 0 0 0 ...
 $ Exits          : num [1:32109] 0 0 0 0 0 0 0 0 0 0 ...
 $ Continent      : chr [1:32109] "OC" "N.America" "N.America" "N.America" ...
 $ Sourcegroup    : chr [1:32109] "(direct)" "(direct)" "Others" "public.tableausoftware.com" ...
 $ Timeinpage     : num [1:32109] 18 4 35 70 81 75 186 710 712 344 ...
 $ Uniquepageviews: num [1:32109] 1 1 1 1 1 1 1 1 1 1 ...
 $ Visits         : num [1:32109] 0 0 0 0 0 0 0 0 1 1 ...
 $ BouncesNew     : num [1:32109] 0 0 0 0 0 0 0 0 0 0 ...
>
```

Data summarization

```
> summary(web_data)
    Bounces            Exits            Continent          Sourcegroup       Timeinpage       Uniquepageviews
 Min.   : 0.000   Min.   : 0.000   Length   :32109   Length   :32109   Min.   :     0.00   Min.   : 1.000
 1st Qu.: 0.000   1st Qu.: 1.000   N.unique :     6   N.unique :     9   1st Qu.:     0.00   1st Qu.: 1.000
 Median : 1.000   Median : 1.000   N.blank  :     0   N.blank  :     0   Median :     0.00   Median : 1.000
 Mean   : 0.713   Mean   : 0.906   Min.nchar:     2   Min.nchar:     4   Mean   :    73.19   Mean   : 1.114
 3rd Qu.: 1.000   3rd Qu.: 1.000   Max.nchar:     9   Max.nchar:    26   3rd Qu.:    10.00   3rd Qu.: 1.000
 Max.   :30.000   Max.   :36.000                                         Max.   :46745.00   Max.   :45.000
     Visits         BouncesNew
 Min.   : 0.000   Min.   :0.00000
 1st Qu.: 1.000   1st Qu.:0.00000
 Median : 1.000   Median :0.01000
 Mean   : 0.906   Mean   :0.00713
 3rd Qu.: 1.000   3rd Qu.:0.01000
 Max.   :45.000   Max.   :0.30000
>
```

Summary for numeric variables



Analyzing whether the unique page view value depends on visits

```
> cor_test_uv <- cor.test(web_data$Uniquepageviews, web_data$Visits, use = "complete.obs")
> print(cor_test_uv)

        Pearson's product-moment correlation

data:  web_data$Uniquepageviews and web_data$Visits
t = 251.51, df = 32107, p-value < 2.2e-16
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.8107300 0.8180958
sample estimates:
      cor
0.8144457
```

The Pearson correlation result shows a strong positive relationship between Uniquepageviews and Visits ($r = 0.8144$, $p < 0.001$). This indicates that as the number of visits increases, the number of unique pageviews also tends to increase. Therefore, unique pageviews appear to depend strongly on visits in this dataset.

Linear regression to quantify the relationship

```
> model_uv <- lm(Uniquepageviews ~ Visits, data = web_data)
> summary(model_uv)

Call:
lm(formula = Uniquepageviews ~ Visits, data = web_data)

Residuals:
    Min      1Q  Median      3Q     Max
-0.1788 -0.1788 -0.1788  0.1353 13.6396

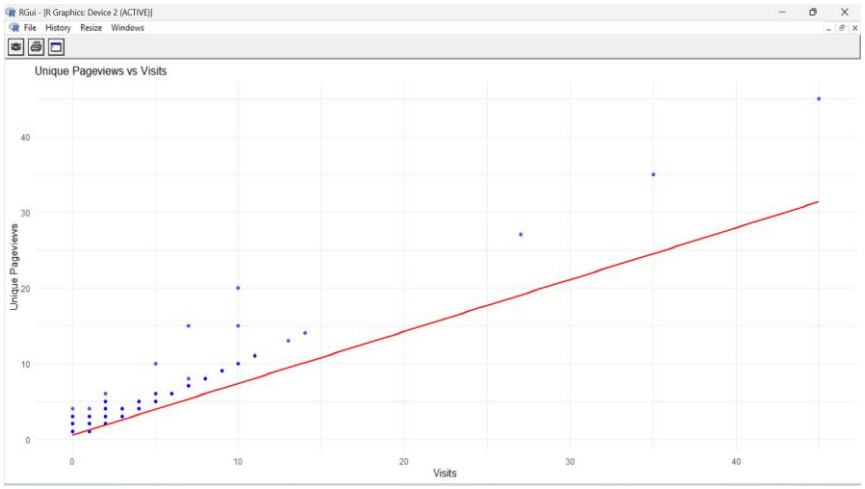
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 0.492837   0.003173   155.3   <2e-16 ***
Visits      0.685945   0.002727   251.5   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3568 on 32107 degrees of freedom
Multiple R-squared:  0.6633,    Adjusted R-squared:  0.6633
F-statistic: 6.326e+04 on 1 and 32107 DF,  p-value: < 2.2e-16

>
```

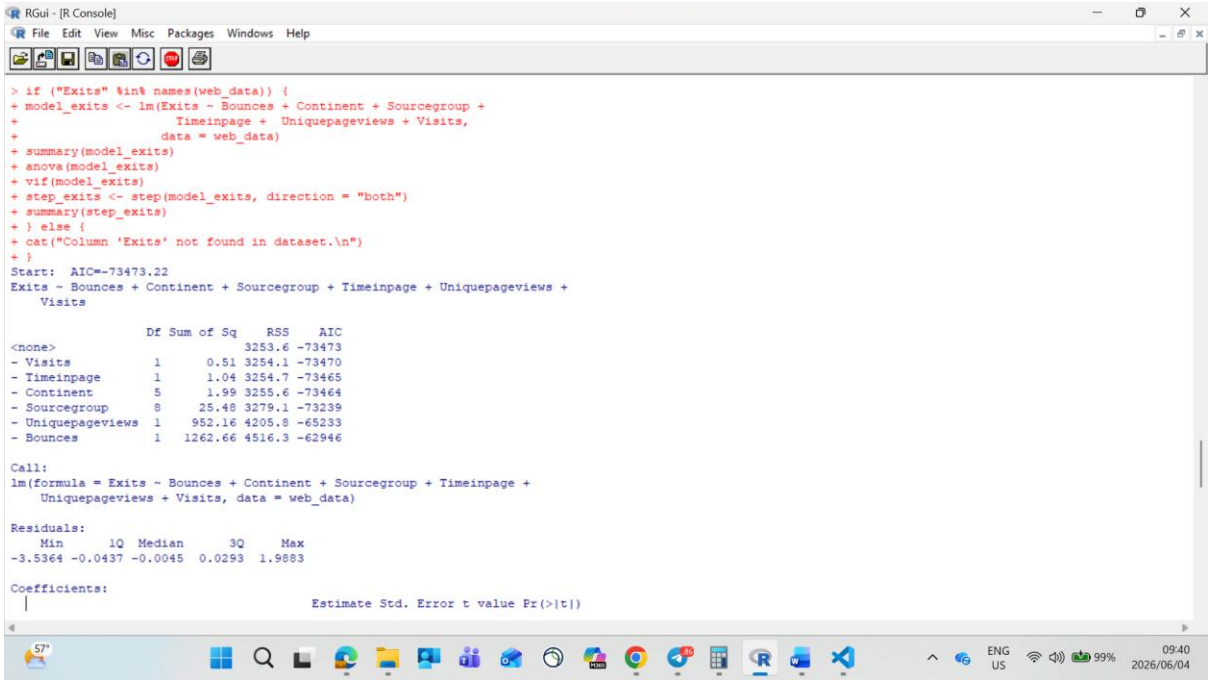
The simple linear regression shows that Visits has a strong positive and statistically significant effect on Uniquepageviews. The coefficient for Visits is 0.6859 with $p < 2e\text{-}16$, indicating that as visits increase, unique pageviews also increase. The model explains about 66.33% of the variation in unique pageviews (R-squared = 0.6633), which suggests a fairly strong relationship between these two variables.

Unique pageviews vs Visits plot



The scatter plot shows a strong positive linear relationship between Visits and Unique Pageviews. As the number of visits increases, unique pageviews also increase, which is confirmed by the upward-sloping regression line. This suggests that Visits is a strong predictor of Unique Pageviews in the dataset.

Factors affecting Exits



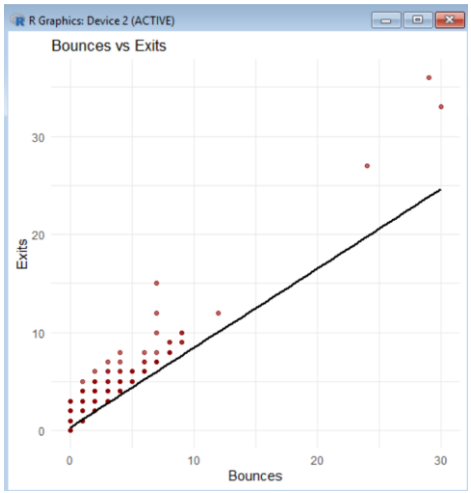
```
Coefficients:
                                          Estimate Std. Error t value Pr(>|t|)
(Intercept)                             -3.223e-02  1.845e-02  -1.747  0.08058 .
Bounces                                  5.132e-01  4.599e-03 111.597  < 2e-16 ***
ContinentAS                              9.433e-03  1.865e-02   0.506  0.61305
ContinentEU                              1.668e-02  1.823e-02   0.915  0.36031
ContinentN.America                      -2.858e-03  1.794e-02  -0.159  0.87341
ContinentOC                              2.562e-03  1.977e-02   0.130  0.89690
ContinentSA                             -1.196e-03  2.127e-02  -0.056  0.95516
Sourcegroupfacebook                      7.563e-02  3.341e-02   2.264  0.02360 *
Sourcegroupgoogle                        3.923e-02  4.769e-03   8.225  < 2e-16 ***
SourcegroupOthers                       -4.697e-03  5.732e-03  -0.819  0.41253
Sourcegroupublic.tableausoftware.com    -5.546e-02  9.542e-03  -5.812 6.22e-09 ***
Sourcegroupreddit.com                    1.170e-02  1.336e-02   0.876  0.38116
Sourcegroupt.co                         -1.429e-02  7.672e-03  -1.863  0.06248 .
Sourcegrouptableausoftware.com          -3.189e-02  7.532e-03  -4.234 2.30e-05 ***
Sourcegroupvisualisingdata.com          -5.212e-02  1.090e-02  -4.782 1.74e-06 ***
Timeinpage                               1.518e-05  4.739e-06   3.203  0.00136 **
Uniquepageviews                          4.943e-01  5.101e-03  96.909  < 2e-16 ***
Visits                                   1.254e-02  5.599e-03   2.239  0.02515 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3184 on 32091 degrees of freedom
Multiple R-squared:  0.7907,    Adjusted R-squared:  0.7906
F-statistic:  7132 on 17 and 32091 DF,  p-value: < 2.2e-16
```

The regression model shows that Exits are strongly influenced by Bounces, Unique Pageviews, Visits, Time on Page, and some Source Group categories. Among all predictors, Bounces and Uniquepageviews are the most important, with highly significant positive effects on Exits ($p < 2e\text{-}16$). This suggests that pages with higher bounce behaviour and more unique pageviews also tend to have more exits.

The model has an R-squared of 0.7907, which means it explains about 79.07% of the variation in Exits. This indicates a very strong model fit. Most of the Continent variables are not statistically significant, meaning continent does not appear to have a strong effect on exits in this dataset. However, several Sourcegroup categories are significant, showing that the way users arrive at the site can influence exit behaviour.

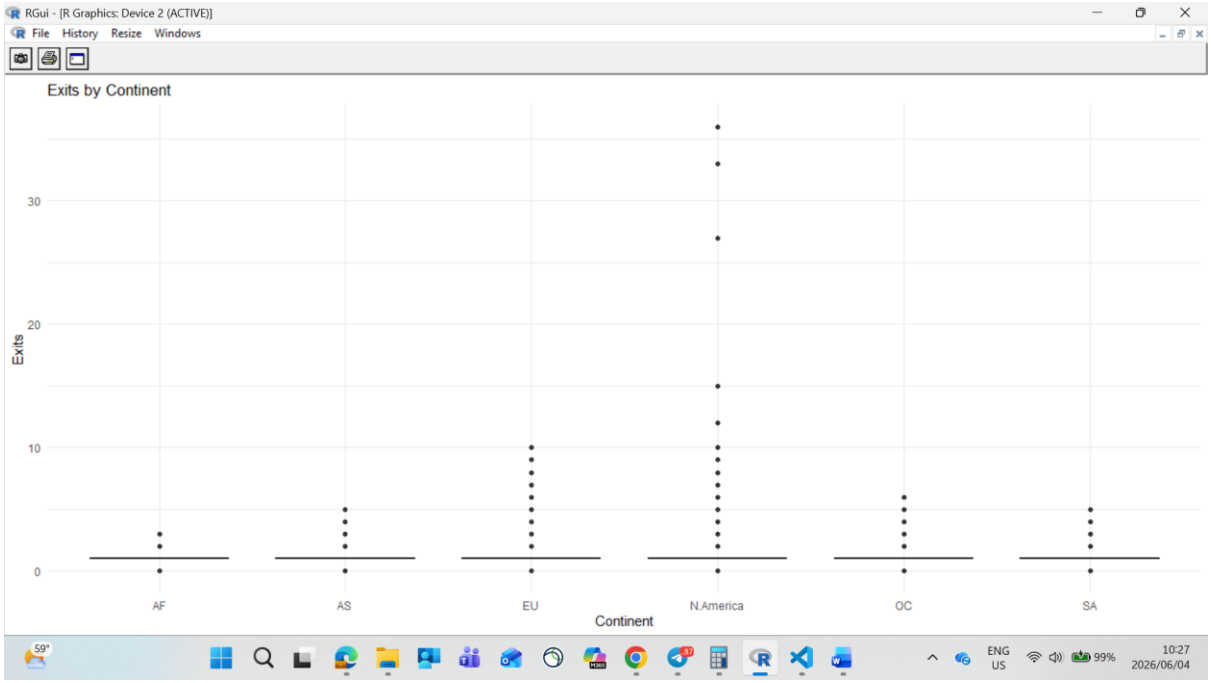
Bounces Vs Exits



As Bounces increase, Exits also increase. The upward-sloping trend line suggests that pages with higher bounce

rates tend to have more exits. This supports the regression result that Bounces is a strong predictor of Exits.

Exits by continent

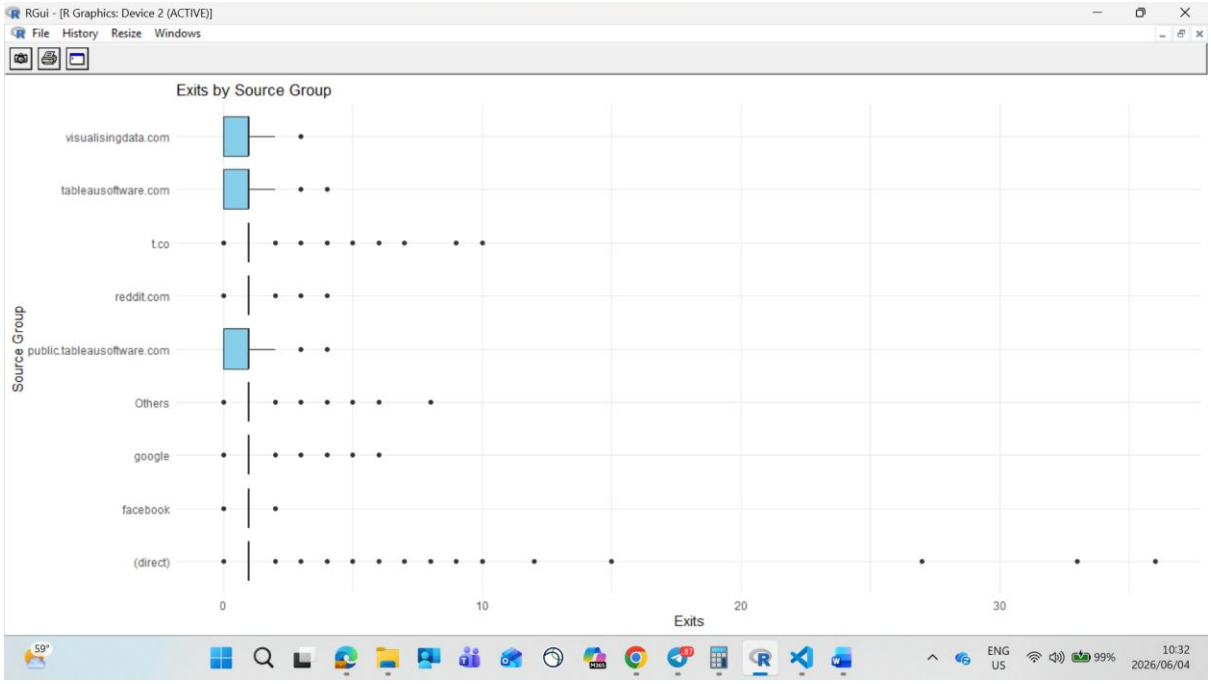


These boxplots suggest that **Exits vary across continents**, but most continents have **low typical exit values** clustered near the bottom.

Observation

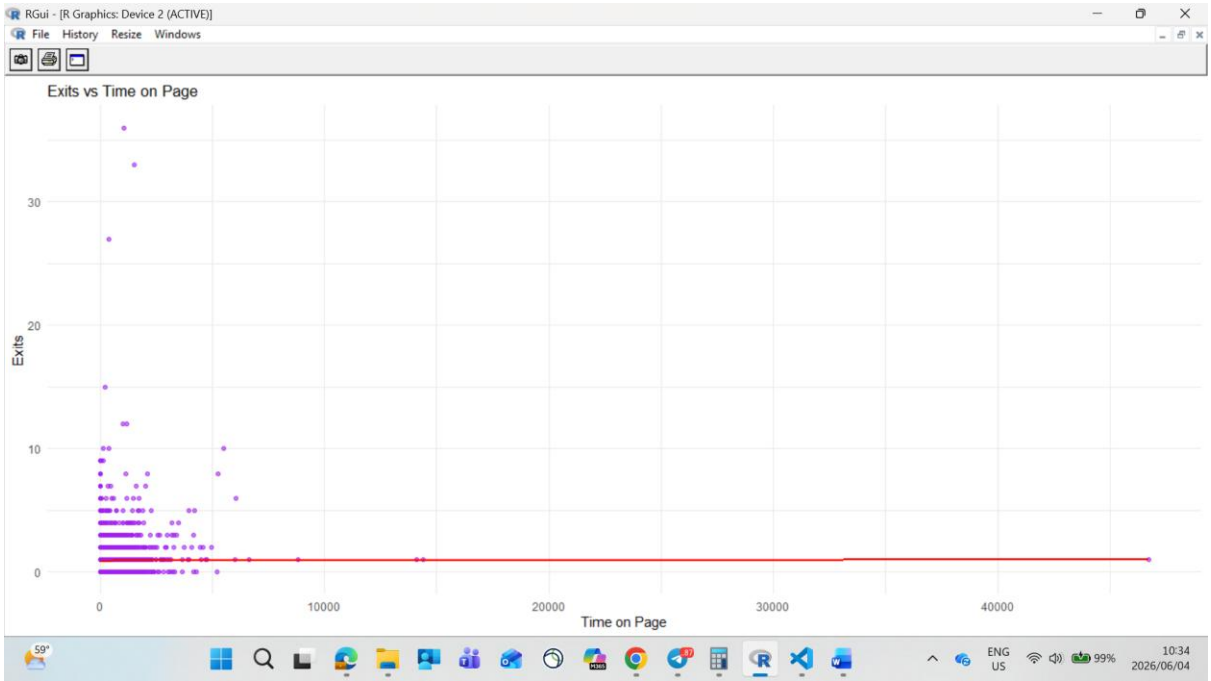
- **N.America** shows the **widest spread** and the **highest exit values**, with several strong outliers.
- **EU** also shows a moderate spread, with exits higher than AF, AS, OC, and SA.
- **AF, AS, OC, and SA** mostly have lower exits, with only a few scattered higher values.

Exits by sourcegroup



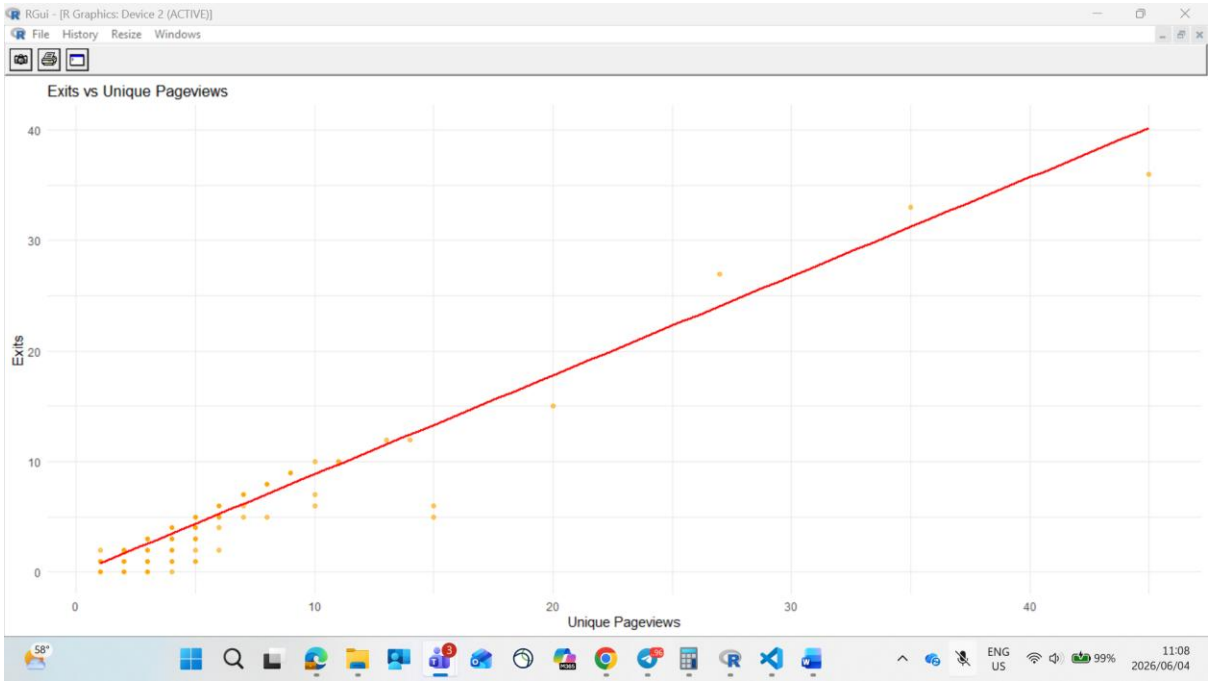
The boxplot shows that exit behaviour differs across source groups. The (direct) source group has the largest spread and several high outliers, indicating that users coming directly to the site sometimes exit at much higher values. t.co and (Others) also show a wider distribution compared with most other sources. In contrast, facebook, google, and tableausoftware.com appear to have relatively lower and more concentrated exit values.

Time on page vs exits



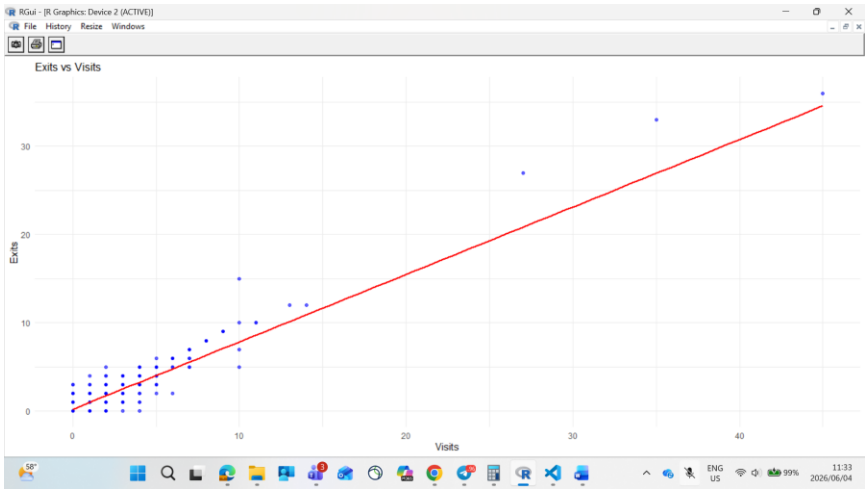
The scatter plot of Exits vs Time on Page shows a weak relationship, with the regression line indicating that time on page has little effect on exits

Exits vs Unique Pageviews



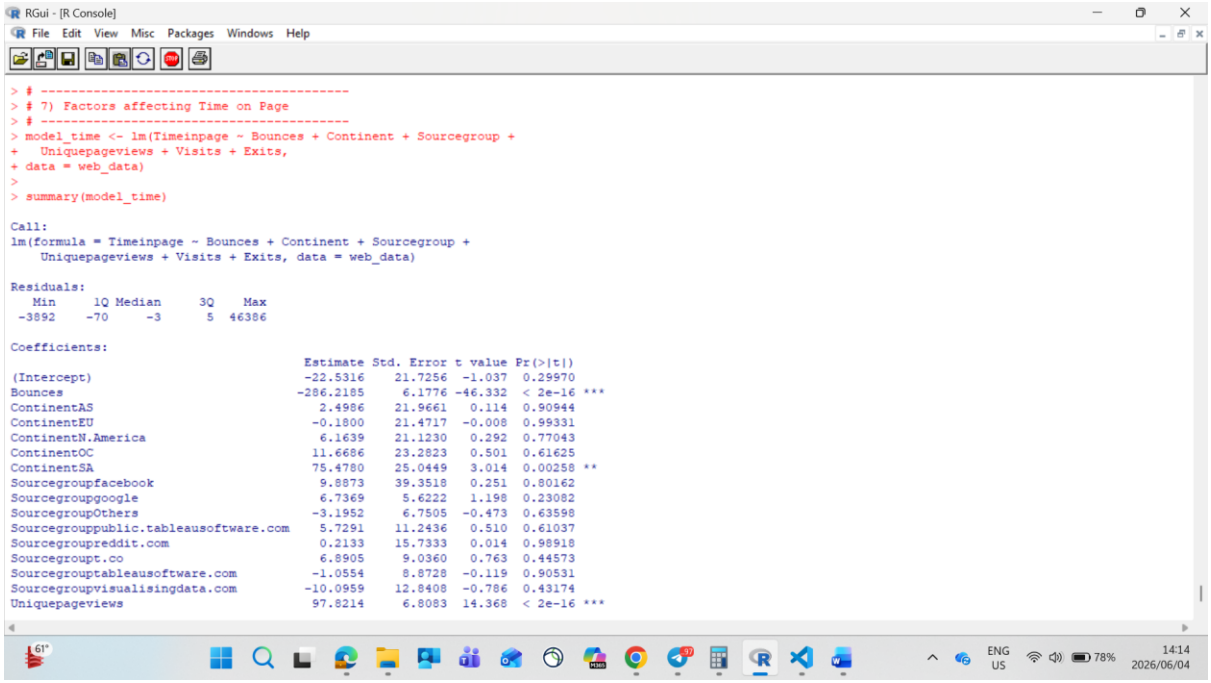
The scatter plot shows a strong positive linear relationship between Unique Pageviews and Exits. As unique pageviews increase, exits also increase, and the upward-sloping regression line confirms this trend. The points are fairly clustered around the line, which suggests that unique pageviews are an important predictor of exits in this dataset.

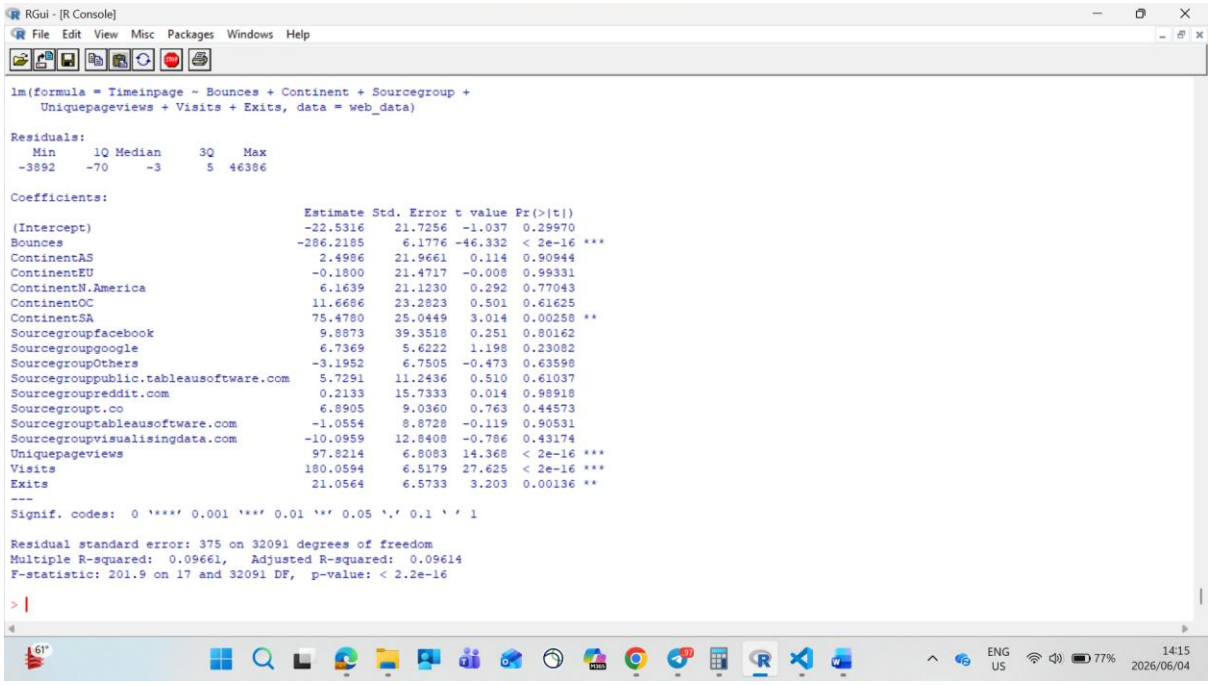
Exits vs Visits



The scatter plot of Exits vs Visits shows a positive relationship, indicating that pages with more visits tend to have higher exits. However, the relationship is weaker than the ones observed for Bounces and Unique Pageviews.

Factors affecting Time on Page

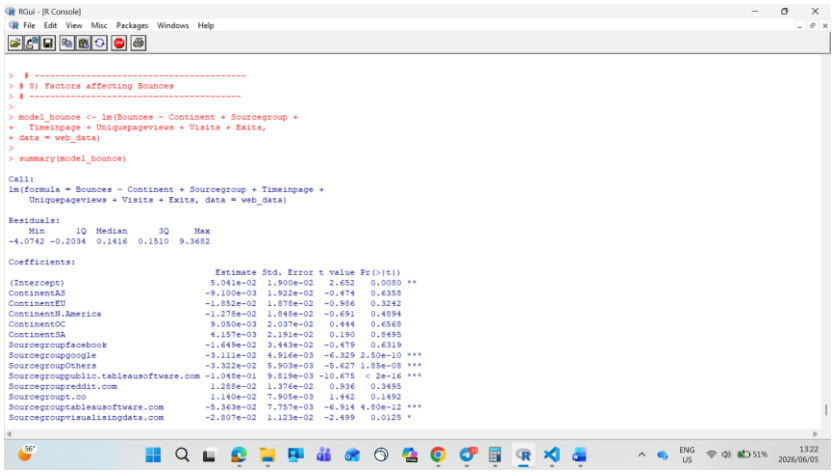


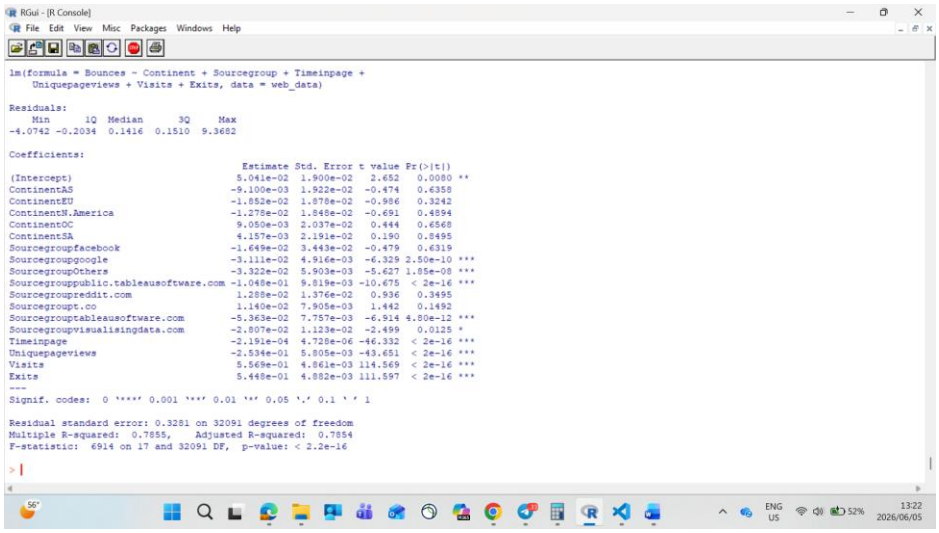


The factors that significantly affect Time on Page are Bounces, Unique Pageviews, Visits, Exits, and South America (SA). Among these, Bounces has the strongest effect, and it is negative (Estimate = -286.2185, $p < 2e\text{-}16$), showing that higher bounce values are

associated with much lower time on page. Of the positive predictors, Visits has the strongest effect (Estimate = 180.0594, p < 2e-16), followed by Unique Pageviews (Estimate = 97.8214, p < 2e-16), South America (SA) (Estimate = 75.4780, p = 0.00258), and Exits (Estimate = 21.0564, p = 0.00136). The model has an R-squared of 0.09661 and an Adjusted R-squared of 0.09614, which means it explains about 9.66% of the variation in Time on Page. This indicates that although some predictors are statistically significant, the overall model fit is relatively weak.

Factors affecting Bounces





The regression model shows that Bounces is significantly affected by Sourcegroup, Time on Page, Unique Pageviews, Visits, and Exits, while most of the Continent variables are not statistically significant. Among the source groups, google, Others, public.tableausoftware.com, tableausoftware.com, and visualisingdata.com all have significant effects on bounce rate, with negative coefficients indicating lower bounce values compared with the reference group. Time on page has a strong negative effect on Bounces (Estimate = -0.0002191, p < 2e-16), showing that higher time on page is

associated with lower bounce rates. Unique Pageviews also has a negative effect (Estimate = -0.2534, $p < 2e\text{-}16$), while Visits (Estimate = 0.5569, $p < 2e\text{-}16$) and Exits (Estimate = 0.5448, $p < 2e\text{-}16$) have the strongest positive effects on Bounces. The model has an R-squared of 0.7855 and an Adjusted R-squared of 0.7854, meaning it explains about 78.55% of the variation in Bounces, which indicates a very strong model fi